



iOS Shortcuts Guide

iOS has powerful home automation tools, and this in-depth guide should give you a headstart. <https://dgit.in/may20-05>



IFTTT applets

Here are a bunch of IFTTT applets to use with your smart lights, including GPS and time based ones. <https://dgit.in/may20-06>

Check if docker was installed properly by running the hello-world image.

```
sudo docker run hello-world

If this produces errors, you must debug them before you continue. Next, install home assistant in docker
curl -sL "https://raw.githubusercontent.com/home-assistant/hassio-installer/master/hassio_install.sh" | sudo
bash -s
```

Now visit your home assistant installation by typing <http://localhost:8123> into your browser. Follow the on screen instructions.

After this, you will need to enable the MQTT integration.

Head to settings -> integrations.

On the bottom right corner of the screen, you should see a button to add integrations. Click on that. On the resulting dialogue box, choose MQTT.

Enter the following settings -

Broker: localhost, Port: 1833, Enable Discovery: True

TUYA-CONVERT

In this step, you will flash Tasmota onto your smart bulb. This step is crucial in stopping your data from leaking all over the internet via the proprietary firmware the bulb already has.

This step needs to be followed on the "flashing" computer. The one with hardware access to its Wi-Fi chip.

First you need to clone the repository from github.

```
sudo apt-get install
-y git

git clone https://
github.com/ct-Open-Source/
Tuya-convert
```

Then you should run the "pre requisites" script. (Note, if you have not already, you should enable the universe repository and update your computer. Otherwise the next step produces errors.)

```
cd Tuya-convert; ./install_prereq.sh
```

Make sure the above step completes without errors. Next, within the Tuya-convert directory execute

```
cat config.txt
```

You should see a line similar to

```
WLAN=wlan0
```

Make sure it matches your Wi-Fi chip. To check the name of your Wi-Fi chip, execute

```
ifconfig | grep wl
```

The first word in the output is the name of your Wi-Fi chip.

```
wlp2s0: flags=4099 mtu
1500
```

Edit config.txt to reflect the right WLAN name. And then finally, you can start the flashing process!

```
./start_flash.sh
```

The following steps are accurate at the time of writing. It is possible that something changes in future updates to Tuya-convert. When in doubt, follow the on screen prompt. Follow the on screen guide, inputting consent as required, until you reach the point where your laptop is broadcasting vtrust-flash (At this point the process pauses for you to press enter). Connect a Wi-Fi device to vtrust-flash. All it needs to do now is connect to Wi-Fi and go to the network login page. Plug your smart bulb into a socket and switch it on. It should immediately start blinking fast. If, for some reason, it isn't blinking fast, you can power cycle it twice to set it to the required mode. To power cycle it, switch it ON-OFF-ON-OFF-ON. Wait a moment for the bulb to enter flashing mode. With your bulb in flashing mode, and a device connected to vtrust-flash, hit enter on the console.

The program will now connect to your smart bulb, make a backup of its original firmware, and then prompt you to choose between installing Tasmota and another firmware. Hit the number choice for Tasmota. The program will upload the Tasmota firmware to your bulb and reboot it. Close the program when it asks you to flash more devices. Now, wait for the bulb to restart, and then connect to its Wi-Fi. It should be named 'something Tasmota-1111'. It is possible to do this and the next configuration steps with your phone browser, but its recommended you do this from your computer.

TASMOTA CONFIGURATION

The first time you connect to your Tasmota device via its Wi-Fi AP, you will need to enter the Wi-Fi name and password of your router. It will then reboot and connect to your Wi-Fi. Once this is done, you will need your bulb's IP address to continue. This can be looked up from your router's status page, or using the "LAN neighbours" features of the "Wi-Fi Analyzer" app on Android. Enter your bulb's IP address into your browser. You will be greeted by Tasmota's webUI menu. Head to Configuration -> Configure MQTT, and enter your MQTT broker's IP. If you've been following this article, that's the same IP address as your "server" computer. Now, hit "Save". Your bulb should reboot again. At this point, your bulb can talk to your MQTT broker. Your MQTT broker is already talking to your install of HASS. Now, all that remains is to tell Tasmota what hardware its running on.

TASMOTA TEMPLATE CONFIGURATION

You now need to configure the template of your firmware to match your device. This is just mapping the software functionality to the hardware ports of your device. Your first try should be to simply copy the template of from your light's page at <https://templates.blakadder.com/bulb.html>.

*Note that in our experiments Wipro Garnet did *NOT* follow the template given on blakadder.*

If the template from the website doesn't give you the expected functionality, and your light is Wipro Garnet NS9001 810lm, you could try the following template:

```
{ "NAME": "WiproSmartBulb"
, "GPIO": [0,0,0,0,38,37,0,0,40,39,41,0,0], "FLAG": 0, "BASE": 18 }
```

If the above steps fail for you, you will have to get your hands dirty and investigate. Its not hard, it just requires some extra time.

STEP 1: Go into console, and type the command `Color 255,0,0,0`